

7th Grade Standard Mathematics

WEEK 1 — DAILY LESSON PLANS

Foundations Review – Statistics & Data Representations

Day	Title	Benchmarks
Day 1	What is a Statistical Question? (Lesson 1.1 Pt 1)	MA.6.DP.1.1
Day 2	Interpreting Histograms & Line Plots (Lesson 1.1 Pt 2)	MA.6.DP.1.4
Day 3	Ratio Relationships (Lesson 1.2 Pt 1)	MA.6.AR.3.1, MA.6.AR.3.2
Day 4	Rates, Unit Rates & Percentages (Lesson 1.2 Pt 2)	MA.6.AR.3.2, MA.6.AR.3.4
Day 5	Rational Numbers on the Number Line (Lesson 1.3 Pt 1)	MA.6.NSO.1.1, MA.6.NSO.1.2

Unit 1: Introduction | Q1 | 26–27 School Year

**DAY
1****What is a Statistical Question?**

Lesson 1.1 — Part 1 | MA.6.DP.1.1

□ Learning Targets

- I can identify statistical questions and explain why they generate numerical data.
- I can distinguish between statistical and non-statistical questions.

□ Warm-Up (8–10 min)**Component: 1.1.1 Warm-Up — Day in the Life of a Mushroom Farmer**

- Play the career profile video (3:46) for the whole class.
- After viewing, students answer 2 reflection questions independently.
- Prompt growth mindset discussion: "How does Alejandro show _____?"
- Consider: adaptability, perseverance, risk-taking.

□ Lesson Activities**Activity 1 | 1.1.2 Refresher: Defining Statistical Questions (10 min)**

- Introduce the definition: A statistical question anticipates variability in the data.
- Review examples: non-statistical vs. statistical questions side by side.
- Use sentence frame: "This is statistical because the answer _____."

Activity 2 | 1.1.3 Try It: Identifying Statistical Questions (15 min)

- Whole-group activity — students volunteer to read each question aloud.
- Poll the class: Is it statistical or not? (thumbs, whiteboards, or hand signals)
- Encourage students to justify their choices before revealing the answer.
- ELL support: Provide written sentence starters for justifications.

□ Practice / Your Turn (8–10 min)**Practice Book — Lesson 1, Questions 1–4**

- Students identify whether each question is statistical or not.
- If not statistical, rewrite to make it statistical.
- Can be completed independently or with a partner.

□ Exit Ticket (3–5 min)**Exit Ticket Prompt:**

- Write one statistical question and one non-statistical question about your school.
- Explain in one sentence WHY each is or is not statistical.

□ Teacher Notes

- Watch the video ahead of time and have the link ready.
- Unit 1 practice book page 1 covers questions 1–4 for this component.
- Extension: Ask students to brainstorm statistical questions they'd want to ask about their own classroom data.

**DAY
2****Interpreting Histograms and Line Plots**

Lesson 1.1 — Part 2 | MA.6.DP.1.4

□ Learning Targets

- I can interpret information from a data set or graphical display.
- I can describe and identify symmetry, skewness, gaps, clusters, outliers, and range.

□ Warm-Up (8–10 min)**Component: 1.1.4 Refresher — Data Displays (8 min)**

- Mini white-board activity: Read off each display type (histogram, line plot, bar graph).
- Students record the letter of the corresponding display on their whiteboard.
- Hold up boards simultaneously — promote whole-class discussion.
- Vocabulary check: histogram, line plot, frequency, interval.

□ Lesson Activities**Activity 1 | 1.1.5 Guided Instruction: Interpreting Histograms and Line Plots (20 min)**

- Introduce the histogram context. Allow students to work Question 1 independently before whole-group discussion.
- Discussion prompts:
 - "What does it mean if something has symmetry? Is this histogram symmetrical?"
 - "What does it mean if data is skewed? What causes skewness?"
 - "Where does most of the data fall? What does that tell us?"
- Partner work for Question 2: compare histogram vs. line plot interpretations.

Activity 2 | 1.1.6 Collaboration: Which Graph? (15 min)

- Partners work together through all parts (1a–1f).
- Begin with a Compare & Contrast Think Aloud: What does each display tell us that the other doesn't?
- Debrief: When would you choose a histogram over a line plot?

□ Practice / Your Turn (8–10 min)**Practice Book — Lesson 1, Questions 5–10**

- Students use the scenario histogram and line plot to answer interpretation questions.
- Questions 6–8: identify number of students surveyed and percentages from the displays.

□ Exit Ticket (3–5 min)**Exit Ticket Prompt:**

- Look at the histogram on the board. Describe TWO features of the distribution (e.g., skewness, outliers, clusters, range).
- Write one sentence explaining what each feature tells you about the data.

□ Teacher Notes

- If students struggle with 'skewed,' have them describe which side looks 'stretched.'
- Connection to MA.7.DP.1.1 (coming in Q2): This lesson builds the vocabulary they'll need for measures of center and variation.

**DAY
3****Ratio Relationships**

Lesson 1.2 — Part 1 | MA.6.AR.3.1, MA.6.AR.3.2

□ Learning Targets

- I can interpret ratio relationships in given contexts.
- I can represent equivalent ratios using tables and double number lines.

□ Warm-Up (8–10 min)**Component: 1.2.1 Warm-Up — Equivalent Fractions (8 min)**

- Students simplify or generate equivalent fractions as mental math warm-up.
- Use as a bridge: "How is finding equivalent fractions related to finding equivalent ratios?"
- ELL support: Use sentence frame — "These ratios are equivalent because _____."

□ Lesson Activities**Activity 1 | 1.2.3 Guided Practice: Representing Ratio Relationships (25 min)**

- Begin with ratio notation review ($a:b$, a to b , a/b).
- Guide students through creating a double number line from a given ratio context.
- Model completing a ratio table — ask students to identify the pattern/rule.
- Real-world context: wins to losses ratio (Tampa Bay Thunder soccer team).
- Pose: "If the team lost 9 games, how many did they win?"
- Discussion: "What does the unit rate mean in this context?"
- ELL sentence frame: "I know the unit rate is _____ because _____."

🗨 Practice / Your Turn (8–10 min)**Practice Book — Lesson 2, Questions 1–4**

- Complete double number line from the Tampa Bay Thunder scenario.
- Questions 2–3: extend the ratio to answer real-world questions.

□ Exit Ticket (3–5 min)**Exit Ticket Prompt:**

- The ratio of cats to dogs at a shelter is 3:5.
- Part A: If there are 15 cats, how many dogs are there? Show using a double number line or table.
- Part B: What is the unit rate of dogs per cat?

□ Teacher Notes

- This is 6th grade review — move at a comfortable pace but keep it purposeful.
- If students finish early: challenge them to write their own ratio word problem using school data.

**DAY
4****Rates, Unit Rates & Percentages**

Lesson 1.2 — Part 2 | MA.6.AR.3.2, MA.6.AR.3.4

□ Learning Targets

- I can find unit rates and use them to solve problems.
- I can apply ratio reasoning to solve percentage problems.

□ Warm-Up (8–10 min)**Warm-Up: Quick Review (5 min)**

- Display: "A car travels 150 miles in 3 hours. What is the unit rate?"
- Cold call or pair-share — connect to yesterday's ratio table work.
- Bridge: "Today we go deeper — rates in distance/speed context and then percentages."

□ Lesson Activities**Activity 1 | 1.2.4 Collaboration: Rates and Unit Rates (20 min)**

- Partners analyze a road trip scenario (Timothy's drive to New Orleans).
- Students compare speeds for different legs of the trip.
- Key question: "Who was driving fastest? How do you know?"
- Extension: "If Timothy drove back at the same speed, how long would it take?"

Activity 2 | 1.2.5 Guided Practice: Applying Ratios to Percentages (15 min)

- Connect ratio reasoning to percent: $\text{percent} = \text{part/whole} \times 100$.
- Context: electronics store sale — items at 70% of original price.
- Students calculate percent discount and find sale prices.
- Scaffold: review the percent bar/model if students need visual support.

□ Practice / Your Turn (8–10 min)**Practice Book — Lesson 2, Questions 5–10**

- Questions 5–6: Rates and speed from the road trip context.
- Questions 7–10: Percent discount and sale price calculations.

□ Exit Ticket (3–5 min)**Exit Ticket Prompt:**

- A jacket originally costs \$80. It is on sale for 25% off.
- Part A: What is the amount of the discount?
- Part B: What is the sale price?
- Show your work using a ratio, equation, or percent model.

□ Teacher Notes

- Common error: students subtract the percent rather than the dollar amount — watch for this.
- Preview: "Tomorrow we start working with ALL rational numbers — including negatives."

**DAY
5****Rational Numbers on the Number Line**

Lesson 1.3 — Part 1 | MA.6.NSO.1.1, MA.6.NSO.1.2

□ Learning Targets

- I can plot, compare, and order rational numbers on a number line.
- I can represent rational numbers in fraction, decimal, and percentage form.

□ Warm-Up (8–10 min)**Component: 1.3.1 Warm-Up — Math Talk (8 min)**

- Display a set of 4 numbers: $-\frac{3}{4}$, 0.6, -1.25, $\frac{3}{5}$
- Ask: "Which is greatest? Which is least? How do you know?"
- Allow think time, then partner discussion, then whole class share.
- Goal: surface prior knowledge and misconceptions about negative numbers and fraction/decimal comparison.

□ Lesson Activities**Activity 1 | 1.3.3 Guided Practice: Plotting Rational Numbers on the Number Line (25 min)**

- Begin with a blank number line — ask students to place 0, 1, and -1.
- Progress: plot fractions → plot decimals → plot negative fractions.
- Key clarification: When given in the same form, compare directly. When in different forms, convert first.
- Guided questions:
 - "Is $-\frac{3}{4}$ closer to 0 or to -1? How do you know?"
 - "How would you place 0.75 on the same number line as $\frac{3}{4}$?"
- Students work in pairs on Part 2 — plotting a mixed set of rational numbers.

Preview: Absolute Value Intro (5 min)

- Introduce the concept: absolute value = distance from zero.
- " $|-4|$ and $|4|$ are both 4 — why?"
- Do not go into full instruction yet — this is a hook for tomorrow (Lesson 1.3 Part 2).

🔗 Practice / Your Turn (8–10 min)**Practice Book — Lesson 3, Questions 1–4**

- Plot and label rational numbers on provided number lines.
- Questions comparing rational numbers using $<$, $>$, $=$ symbols.

□ Exit Ticket (3–5 min)**Exit Ticket Prompt:**

- Plot these four numbers on the number line provided: -2, $\frac{1}{2}$, -0.75, 1.5
- Then write them in order from least to greatest.

□ Teacher Notes

- Watch for: students placing negative fractions on the positive side.
- Watch for: students thinking $-0.75 > -0.25$ because $75 > 25$ — the 'larger negative' misconception.
- Week 2 preview: next week we complete Lesson 1.3 (absolute value) and move through Lessons 1.4–1.6.

